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Project Title

FindItNow – Real-Time Lost & Found Management System Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Lost items are a common problem in public places such as colleges, hostels, offices, transport hubs, and events. Most lost-and-found systems today are manual, inefficient, and poorly organized, relying on notice boards, word of mouth, or social media posts. These approaches make it difficult to search for items in real time and often result in items never being reunited with their owners.

The motivation behind this project is to design and implement a **real-time digital Lost & Found system** that allows users to register, report lost items with detailed information such as date, time, location, image, and description, and search for matching found items efficiently. The system also encourages honesty and social responsibility by rewarding users who help return lost items with goodwill points.

From a DAA perspective, this project is important because it applies efficient data structures and algorithms for searching, matching, and managing large datasets in real time. The project demonstrates how algorithmic design directly impacts usability, performance, and scalability in real-world applications.

State of the Art / Current Solutions

Currently, most lost-and-found systems are either manual registers or simple web forms with limited search functionality. Some mobile apps and websites exist, but many lack efficient matching algorithms, verification workflows, or incentive mechanisms. Social media platforms are also used, but they are unstructured, unreliable, and inefficient for systematic searching.

Existing solutions often fail to scale well when the number of reported items grows, as searching is mostly keyword-based without optimized algorithms. Verification of ownership is also weak, leading to false claims or misuse. Our proposed system improves upon these limitations by integrating algorithm-driven search, structured data storage, verification workflows, and optional automation.

Project Goals and Milestones

Goals

- Design a real-time Lost & Found system using C++ concepts and DAA principles.
- Enable users to register accounts and securely manage lost or found item reports.
- Implement efficient searching and matching algorithms based on multiple attributes.
- Introduce an admin or automated verification workflow for authenticity.

- Reward users who help recover lost items with goodwill points.

Milestones

- Requirement analysis and system design (Week 1)
 - Design data structures and algorithms (Week 2)
 - Implement user registration and authentication (Week 3)
 - Implement lost item reporting and image handling (Week 4)
 - Implement search and matching algorithms (Week 5)
 - Admin/automated verification and reward system (Week 6)
 - Testing, optimization, and final documentation (Week 7–8)
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Project Approach

The project will be developed primarily in **C++**, focusing on DAA concepts. Data such as users, items, and transactions will be stored using optimized data structures like hash tables, trees, and vectors.

Key approaches include: - Using **hashing** for fast user account lookup and item indexing. - Applying **searching and filtering algorithms** based on keywords, date, location, and item category. - Using **sorting algorithms** to rank search results by relevance or time. - Implementing an **admin verification module** or an optional automated workflow based on confidence scores. - Designing a modular system that separates user, admin, and item management logic.

The system will initially be console-based, with scope for future extension to GUI or web interfaces.

System Architecture (High Level)

Components: - User Interface (User registration, item reporting, searching) - Lost Item Management Module - Found Item Matching Engine (Algorithms) - Admin Verification / Automated Workflow Module - Reward & Reputation System - Data Storage Layer

Flow: User → Login/Register → Report/Search Item → Matching Algorithm → Admin/Auto Verification → Item Marked Found → Reward Assigned

Project Outcome / Deliverables

- A fully functional C++ Lost & Found system.
- Efficient implementation of DAA concepts such as hashing, searching, and sorting.
- Admin and automated verification workflows.
- Reward-based goodwill point system.
- Complete project report with design, algorithms, and analysis.
- Demonstration of real-time searching and matching.

Assumptions

- Users provide correct and honest information while reporting items.
 - The system is initially used in a controlled environment (college or campus).
 - Images are stored as file references or paths.
 - Admin access is trusted and secure.
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References

- Geeks for Geeks
- DAA by Dr Anupam Singh (GEHU)
- DSA by Dr Hriday Kr Gupta (GEHU and You tube)